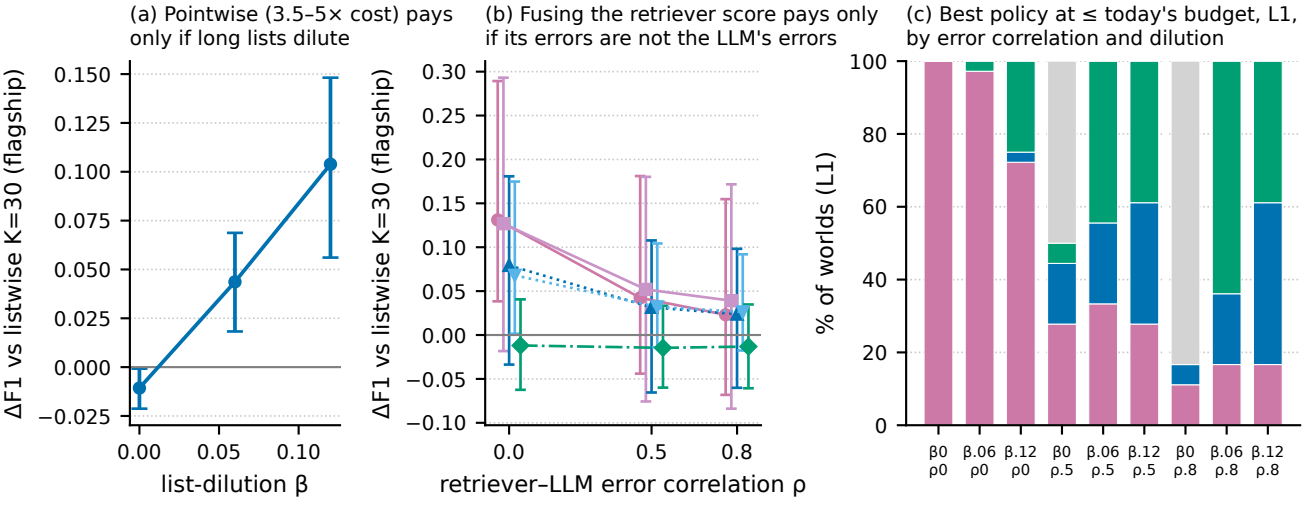
(a) Pointwise (3.5–5× cost) pays only if long lists dilute
ΔF1 vs listwise K=30 (flagship)
0.150
0.125
0.100
0.075
0.050
0.025
0.000
−0.025
0.00
0.05
0.10
list-dilution β
(b) Fusing the retriever score pays only if its errors are not the LLM's errors
ΔF1 vs listwise K=30 (flagship)
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
0.0
0.5
0.8
retriever–LLM error correlation ρ
(c) Best policy at ≤ today's budget, L1, by error correlation and dilution
% of worlds (L1)
100
80
60
40
20
0
β0 ρ0
β.06 ρ0
β.12 ρ0
β0 ρ.5
β.06 ρ.5
β.12 ρ.5
β0 ρ.8
β.06 ρ.8
β.12 ρ.8
LW K=30 + retriever prior (1.0× cost)
LW K=10 + retriever prior (0.47×)
LW K=10 (0.47×)
retriever-prune (0.67×)
cascade band 0.4 (0.57×)
fusion: + retriever prior (any K)
read less: K=10 / prune / adaptive-K
two models: cascade / route
today's design (flagship K=30)